

FOR LOOP

- * Write a program to find the sum of all even numbers between 1 and n.
- * Write a program to find the sum of all numbers between 1 and n that are divisible by 3.
- * Write a program to print all numbers between 1 and n that are divisible by both 3 and 5.
- * Write a program to print all prime numbers between 1 and n.
- * Write a program to find the sum of prime numbers between 1 and n.
- * Write a program to print all factors of a given number.
- * Write a program to check whether a given number is a perfect number.
- * Write a program to calculate the factorial of a number using only a for loop.
- * Write a program to print the multiplication table of a number from 1 to 10.
- * Write a program to find the second largest number among n input numbers without sorting.

WHILE LOOP

- * Write a program to count the number of digits in a given number.
- * Write a program to find the sum of digits of a number.
- * Write a program to reverse a number using while loop.
- * Write a program to check whether a number is a palindrome.
- * Write a program to find all factors of a number using while loop.
- * Write a program to find the largest digit in a number.
- * Write a program to find the smallest digit in a number.
- * Write a program to find the frequency of a given digit in a number.
- * Write a program to repeatedly calculate the sum of digits until a single-digit number is obtained.
- * Write a program to perform a circular shift of a number by one digit.

NUMBER LOGIC

- * Write a program to reverse a number without converting it into a string.
- * Write a program to check whether a number is a palindrome without using strings.
- * Write a program to check whether a number is an Armstrong number.
- * Write a program to find the frequency of every digit present in a number.
- * Write a program to find the second largest digit of a number without sorting.
- * Write a program to find the sum of digits at even positions and odd positions separately.
- * Write a program to check whether all digits of a number are even.
- * Write a program to check whether the digits of a number are in increasing order.
- * Write a program to find the difference between the largest and smallest digit of a number.
- * Write a program to find the largest and second-largest digits in a single traversal of the number.

PATTERN PRINTING

Print the following pattern using nested for loops:

```
*  
**  
***  
****  
*****
```

Print the following inverted pattern using nested for loops:

```
*****  
****  
***  
**  
*
```

Print a hollow right-angled triangle:

```
*  
**  
* *  
* *  
*****
```

Print the following number pattern:

```
1  
12  
123  
1234  
12345
```

Print the following number pattern:

```
1  
22  
333  
4444  
55555
```

Print the following number pattern:

```
12345  
1234  
123  
12  
1
```

Print the following centered number pattern:

```
  1
 121
12321
1234321
123454321
```

Print a hollow square with both diagonals:

```
*****
**  **
*    *
**  **
*****
```

Print the following X pattern:

```
*  *
*  *
*
*  *
*  *
```

Print a normal diamond pattern using stars:

```
  *
 ***
*****
*****
*****
*****
*****
***
  *
```

Print a hollow diamond pattern using stars:

```
  *
 * *
*   *
*   *
*     *
*     *
*   *
*   *
*   *
  *
```

Print a butterfly pattern using nested for loops:

```
*      *
**     **
***    ***
****   ****
*****
*****
****   ****
***    ***
**     **
*      *
```